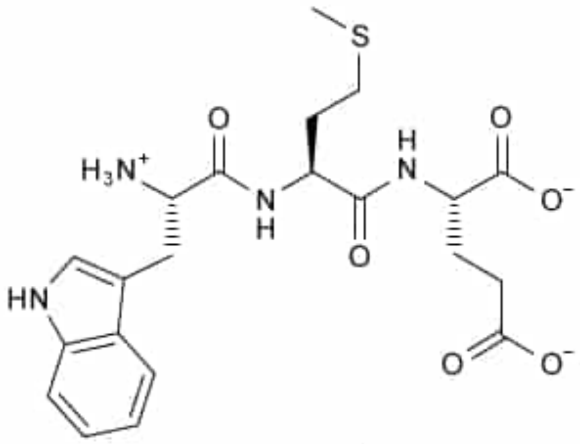
Compound **X** is shown below.



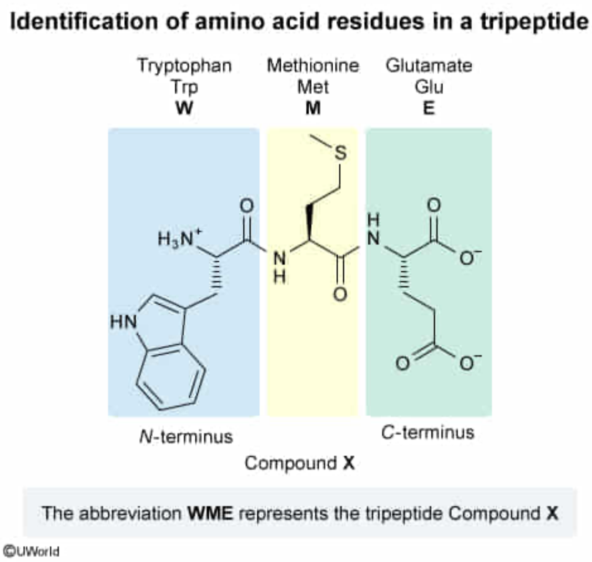
Compound **X**

Which of the following corresponds to the tripeptide Compound **X**?

- A. HCD [3%]
- B. HME [10%]
- C. WME [67%]
- D. WMD [18%]

Correct
67% answered correctly

Explanation:



The structure of an **amino acid** consists of a carboxyl group, an amino group, and a side chain, all bonded to the same carbon atom (the α-carbon). The main difference in the structure of the 20 common amino acids is the side chain, which has different functional groups. Amino acids are classified by the nature of the side chains (eg, hydrophilic, hydrophobic, basic, acidic) and can be named using shorthand **one- and**

three-letter abbreviations. Polypeptide names list the amino acid residues from the end containing the free α-amino group (the *N*-terminus) to the end containing the free carboxyl group (the *C*-terminus).

In this question, Compound **X** is a tripeptide, which has three amino acid residues. The *N*-terminus is on the left side of the structure, and the *C*-terminus is on the right side. From the *N*-terminus terminus, the first residue contains an indole group and is tryptophan (Trp, W). The second residue has a side chain containing a thioether group and is methionine (Met, M). The residue at the *C*-terminus has a side chain containing a carboxylate group and a total of three carbon atoms (including the carboxylate group) and is glutamate (Glu, E).

Therefore, the abbreviation that corresponds to Compound **X** is **WME**.

(Choice A) HCD corresponds to a tripeptide containing the residues histidine (His, H), whose side chain contains an imidazole; cysteine (Cys, C), whose side chain contains a thiol; and aspartate (Asp, D), which contains a carboxylate side chain. These residues are *similar* (but *not* the same) to those in Compound **X**.

(Choice B) The first residue in HME is incorrect. H corresponds to histidine, the amino acid with a heterocyclic imidazole group; however, the first residue in Compound **X** is tryptophan (W), which contains a heterocyclic indole group.

(Choice D) The final residue in WMD is incorrect. D corresponds to aspartate, whose side chain contains one less CH_2 group than glutamate (E), the correct final residue in Compound **X**.

Educational objective:
Amino acids contain a carboxyl group, an amino group, and a side chain, all bonded to the same carbon atom. The structure of amino acids differs mainly with the functional groups contained in the side chain. In addition to their chemical names, amino acids have one- and three-letter abbreviations.

Time spent:0 sec | **QID**:404116
Subject: Organic Chemistry | **System:** Functional Groups and Their Reactions